BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 067 · Lesson 067 · Divide and Conquer: Splitting the Search

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is decomposition?

Q2. Which input assumption does binary search require?

Q3. How does Math.Min(a,b) behave?

Q4. Binary search [2,4,6,8,10,12,14] for 12: show low/high and midpoint indexes.

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I Do · Read new knowledge

Key vocabulary

divide and conquer — Solve smaller subproblems and combine.

precondition — Requirement for correctness.

binary search — Halving search on sorted data.

merge — Combine sorted partial results.

New learning

Define and predict — Divide and conquer breaks a problem into smaller instances, solves them, and (where relevant) combines results.

Explain — Binary search repeatedly discards half of a SORTED array. It is an example of reducing to one subproblem rather than solving both.

Evaluate — Merge sort divides into halves and combines them in sorted order. Dividing alone is not enough; correctness depends on valid preconditions and combining.

Worked example

Search sorted [2,5,8,11,14,17,20,23] for 20.

1. Follow the reasoning — Each iteration narrows the candidate interval, and result is index 6.

We Do · Practise with guidance

Search sorted [3,6,9,12,15,18,21,24] for 6.

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mid3=12 → left indexes0–2; mid1=6 → found at index1.
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Follow the reasoning

The second comparison finds the target after removing upper half.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

For [1,4,7,10,13,16,19,22] trace middle indexes when target19. [6]

Question 2

Explain divide, solve and combine for merge sort on [7,2,9,1]. [6]

Question 3

Compare repeated halving with linear scan for sorted list of 1024 items, in terms of worst-case comparisons and preconditions. [6]

Exit Ticket · Answer before checking

Binary search will correctly find any target in any unsorted list.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Breaking a task into smaller independent parts.

Q2. Sorted elements.

Q3. Returns the smaller argument.

Q4. low0 high6 mid3=8, then low4 high6 mid5=12; found index5.

You Do pages · Independent questions

Q1. mid3=10 [2], low4 then mid5=16 [2], low6 mid6=19 [2].

Q2. Split [7,2] and [9,1] [2], solve [2,7] and [1,9] [2], merge [1,2,7,9] [2].

Q3. Linear O(n), up to1024 checks [2]; binary O(log n), at most ~11 candidate checks with conventional bounds [2]; sorted data required [2].

Exit Ticket · Model answer

False. Sorting is required.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: